Complete cold DLP runtime through ecbench

Run 20261019-190253-boolean-adaptive-ecbench-cbeb18, source 9bed104b0a. Complete one-target cold DLP recovery on dedicated c7i.2xlarge; native ecbench L2 requirement, one worker,8 independent public targets per curve,5 measured rounds plus1 warmup,11 interleaved arms. Sources/specs and complete audit certificates remain adjacent/raw. Missing, failed and below-level rows are retained; a launch is not a result.

All methods within each session share the same public targets and standard polynomial subspace: dimension5 at n17 and6 at n23. These bases differ from the component experiment's Frobenius-invariant bases. The curve identities are icv1-f2m17-tm101-00378d4e and icv1-f2m23-tm5197-1f85e9e1. This two-size panel is not a scaling claim.

The pipeline solve clock includes factor-base construction, oracle prepare, relation collection, linear algebra, target descent/recovery and internal verification. Whole child process additionally includes adapter construction, report metadata, common workload construction and startup. Both clocks are recorded; the native harness's comparison gate uses the pipeline solve clock. Independent parent verification/ replay, compilation and warmup are outside these clocks.

F4/F5 baseline uses the prior recursive policy with qualified GF2 kernels. Sources0acfc55b7f/71c4c99d85 use candidate matrix_runtime=quadratic-tail; corrected source9bed104b0a uses adaptive-quadratic-tail, with a different method identity. F6 names kernel_runtime=native-v1. Generalized typed roots and both Semaev table backends participate. Environment-override refusal remains intact; no admission threshold is relaxed.

Wall ratios below are B/A from ecbench's own paired bootstrap and A/A gate. Admission, completeness and outside_noise must all support a gain; partial/failed sessions do not become successful comparisons. Process medians are supplementary observed clocks, not a replacement gate.

Counted S, generic floor and matched strong rho are retained in table-n*.txt and ecbench-summary.json. Gray assignments are explicitly counted unpriced; reference native unpriced counters are retained too. Sensitivity bands at0/0.1/1 GAE per recorded operation are diagnostic and deduplicate generalized solver-count aliases. An unquantified counter stays unquantified. No ratio of lower bounds is a full-cost ranking.

Exact original-output and replay checks, query-count comparisons, source/binary hashes, failures and job resources remain in the evidence. All-L2 coverage is separate from execution/verification completeness; a fully verified session can have below-level timing rows. Low-sample clocks are descriptive; no threshold is relaxed. New matrix policies stay opt-in; no external-fastest, other-hardware or rho-beating claim is made.

n17: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

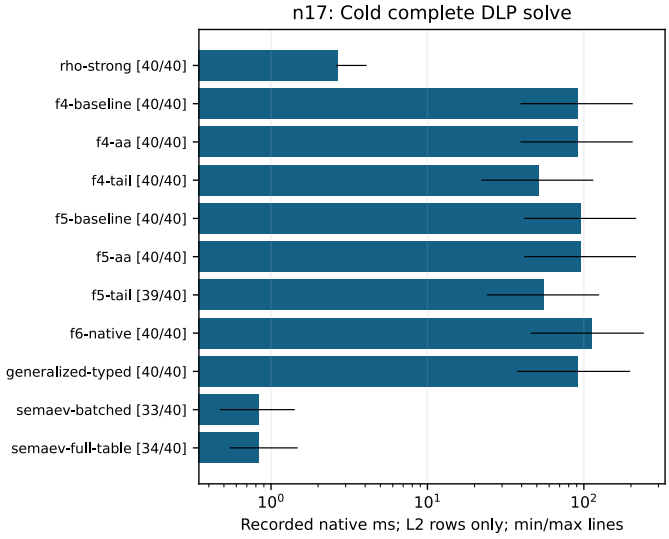
n17 f4: candidate/baseline pipeline wall 0.5623, CI95 [0.5599368512706958, 0.5651490815056114], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

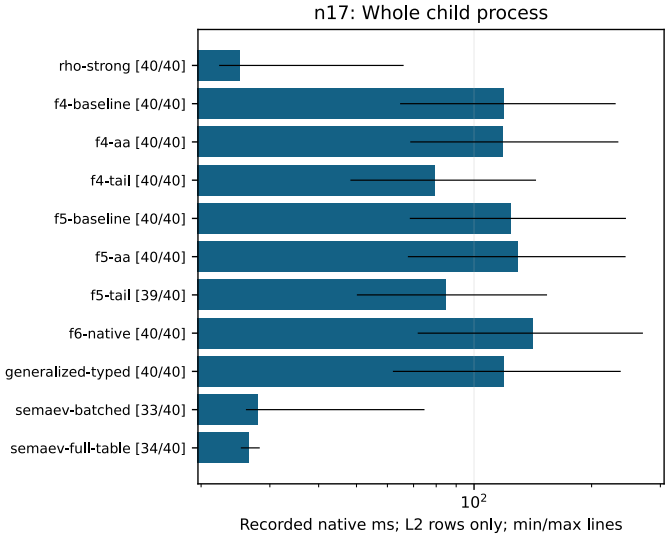
n17 f5: candidate/baseline pipeline wall 0.5822, CI95 [0.5795548493574879, 0.5839443622722156], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

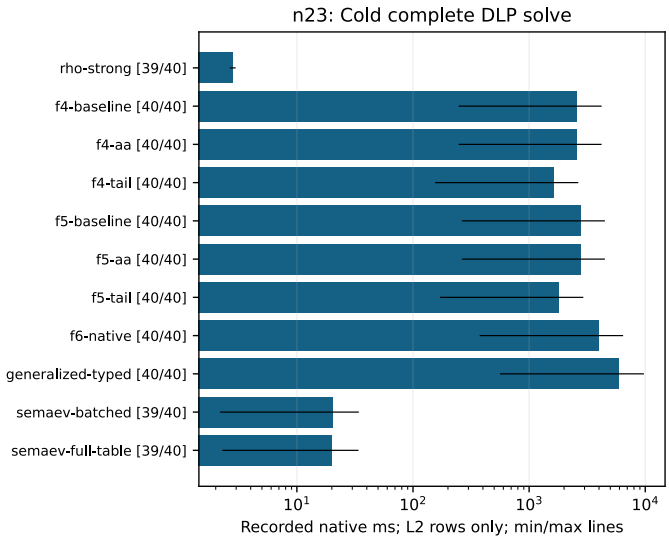
n23: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

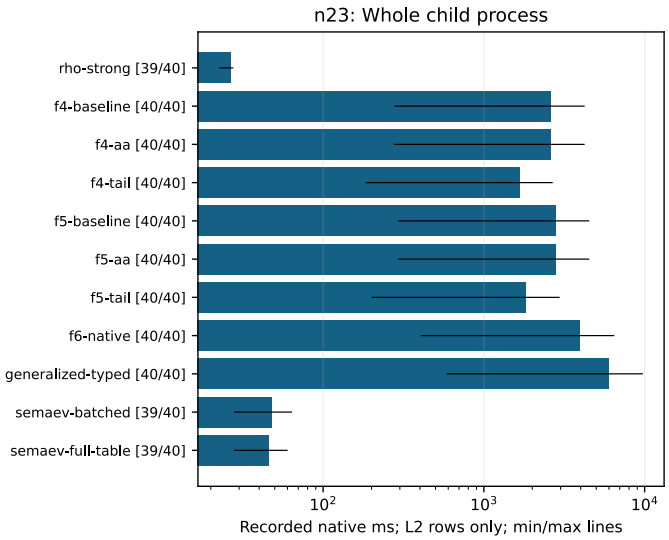
n23 f4: candidate/baseline pipeline wall 0.6309, CI95 [0.6299009971212228, 0.6315614249975599], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

n23 f5: candidate/baseline pipeline wall 0.6532, CI95 [0.6526133456284446, 0.6539370178846236], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.


n17: Cold complete DLP solve
rho-strong [40/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [39/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [33/40]
semaev-full-table [34/40]
10^0
10^1
10^2
Recorded native ms; L2 rows only; min/max lines


n17: Whole child process
rho-strong [40/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [39/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [33/40]
semaev-full-table [34/40]
10^2
Recorded native ms; L2 rows only; min/max lines


n23: Cold complete DLP solve
rho-strong [39/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [39/40]
semaev-full-table [39/40]
10^1
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines


n23: Whole child process
rho-strong [39/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [39/40]
semaev-full-table [39/40]
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines

Recorded clocks and admission

n	Arm	Solve ms	Process ms	Verified	L2	Requested	All L2
17	rho-strong	2.6819	25.1376	40	40	40	True
17	f4-baseline	91.3372	119.0197	40	40	40	True
17	f4-aa	91.2443	117.9840	40	40	40	True
17	f4-tail	51.3415	79.3053	40	40	40	True
17	f5-baseline	95.9779	124.2733	40	40	40	True
17	f5-aa	95.8601	129.1508	40	40	40	True
17	f5-tail	55.9904	84.5148	40	39	40	False
17	f6-native	113.0123	141.3677	40	40	40	True
17	generalized-typed	91.9735	118.9833	40	40	40	True
17	semaev-batched	0.8323	27.9561	40	33	40	False
17	semaev-full-table	0.8427	26.4699	40	34	40	False
23	rho-strong	2.7805	26.6863	40	39	40	False
23	f4-baseline	2592.6712	2619.9191	40	40	40	True
23	f4-aa	2593.9296	2623.7207	40	40	40	True
23	f4-tail	1636.2769	1666.0858	40	40	40	True
23	f5-baseline	2764.2312	2793.9766	40	40	40	True
23	f5-aa	2766.4568	2796.0575	40	40	40	True
23	f5-tail	1806.6459	1836.5007	40	40	40	True
23	f6-native	3952.1317	3981.7135	40	40	40	True
23	generalized-typed	5946.1759	5972.4103	40	40	40	True
23	semaev-batched	20.5768	47.8789	40	39	40	False
23	semaev-full-table	19.9445	46.3640	40	39	40	False

Native harness comparisons; B/A

n	Pair	Wall gate	Median ratio	CI95	Outside noise	Below L2
17	f4	admitted	0.5623	0.5599..0.5651	True	0
17	f5	admitted	0.5822	0.5796..0.5839	True	1
17	rho-f4-tail	admitted	19.1779	17.2474..22.8799	True	0
17	rho-f5-tail	admitted	20.9193	19.1648..25.4592	True	1
17	rho-f6-native	admitted	41.3187	36.7152..48.1579	True	0
17	rho-generalized-typed	admitted	34.2003	30.2126..39.1385	True	0
17	rho-semaev-batched	admitted	0.3056	0.2803..0.3438	True	7
17	rho-semaev-full-table	admitted	0.3111	0.2959..0.3514	True	6
23	f4	admitted	0.6309	0.6299..0.6316	True	0
23	f5	admitted	0.6532	0.6526..0.6539	True	0
23	rho-f4-tail	admitted	586.4033	419.1506..692.5298	True	1
23	rho-f5-tail	admitted	647.2587	462.0887..762.5880	True	1
23	rho-f6-native	admitted	1416.4945	1014.6719..1680.5328	True	1
23	rho-generalized-typed	admitted	2137.0665	1525.7469..2506.3016	True	1
23	rho-semaev-batched	admitted	7.3576	5.6803..8.4332	True	2
23	rho-semaev-full-table	admitted	7.2811	5.5185..8.3262	True	2

Partly priced S sensitivity; all entries diagnostic

n	Arm	0 GAE/op	0.1 GAE/op	1 GAE/op	Lower bound	Unknown entries
17	rho-strong	4.0416	4.185	5.4753	True	0
17	f4-baseline	43.799	43.799	43.799	False	0
17	f4-aa	43.799	43.799	43.799	False	0
17	f4-tail	33.227	33.288	33.842	True	0
17	f5-baseline	43.731	43.731	43.731	False	0
17	f5-aa	43.731	43.731	43.731	False	0
17	f5-tail	33.164	33.225	33.779	True	0
17	f6-native	33.082	33.082	33.082	False	0
17	generalized-typed	16.421	1080.1	10653	True	0
17	semaev-batched	17.317	17.317	17.317	False	0
17	semaev-full-table	18.083	18.083	18.083	False	0
23	rho-strong	0.88552	0.95031	1.5335	True	0
23	f4-baseline	72.251	72.251	72.251	False	0
23	f4-aa	72.251	72.251	72.251	False	0
23	f4-tail	65.998	66.034	66.354	True	0
23	f5-baseline	72.228	72.228	72.228	False	0
23	f5-aa	72.228	72.228	72.228	False	0
23	f5-tail	65.97	66.006	66.326	True	0
23	f6-native	65.541	65.541	65.541	False	0
23	generalized-typed	43.007	16897	1.6859e+05	True	0
23	semaev-batched	43.709	43.709	43.709	False	0
23	semaev-full-table	44.039	44.039	44.039	False	0

Explicit unpriced operation ledgers; different units are not equated

n	Arm	Unit	Samples	Median count	Min	Max
17	rho-strong	canonicalisations_uncharged	40	192.0	64	356
17	rho-strong	partition_hashes_uncharged	40	148.5	30	306
17	rho-strong	table_inserts_uncharged	40	9.0	2	20
17	rho-strong	table_queries_uncharged	40	10.0	3	22
17	f4-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	f5-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	generalized-typed	generalized_assignments_uncharged	40	2613760.0	1051648	5709824
23	rho-strong	canonicalisations_uncharged	40	672.0	160	1217
23	rho-strong	partition_hashes_uncharged	40	594.5	119	1105
23	rho-strong	table_inserts_uncharged	40	39.5	12	79
23	rho-strong	table_queries_uncharged	40	40.5	13	80
23	f4-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	f5-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	generalized-typed	generalized_assignments_uncharged	40	355270656.0	33546240	580587520

Admitted phase clocks; recorded ms

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
17	f4-baseline	factor_base	0.1368	0.1314	0.1656	40
17	f4-baseline	linear_algebra	0.0111	0.0040	0.0211	40
17	f4-baseline	oracle_setup	0.0489	0.0457	0.0659	40
17	f4-baseline	relations	91.0934	39.1054	205.1595	40
17	f4-baseline	verify	0.0021	0.0016	0.0031	40
17	f4-aa	factor_base	0.1370	0.1307	0.1603	40
17	f4-aa	linear_algebra	0.0099	0.0054	0.0204	40
17	f4-aa	oracle_setup	0.0487	0.0444	0.1227	40
17	f4-aa	relations	90.9996	39.0758	204.9668	40
17	f4-aa	verify	0.0020	0.0016	0.0027	40
17	f4-tail	factor_base	0.1362	0.1303	0.1634	40
17	f4-tail	linear_algebra	0.0107	0.0022	0.0158	40
17	f4-tail	oracle_setup	0.0490	0.0436	0.0565	40
17	f4-tail	relations	51.0986	21.8000	114.7090	40
17	f4-tail	verify	0.0020	0.0016	0.0025	40
17	f5-baseline	factor_base	0.1356	0.1290	0.1634	40
17	f5-baseline	linear_algebra	0.0094	0.0023	0.0187	40
17	f5-baseline	oracle_setup	0.0472	0.0438	0.0582	40
17	f5-baseline	relations	95.7273	41.3605	215.7969	40
17	f5-baseline	verify	0.0020	0.0016	0.0026	40
17	f5-aa	factor_base	0.1356	0.1310	0.1606	40
17	f5-aa	linear_algebra	0.0103	0.0022	0.0169	40
17	f5-aa	oracle_setup	0.0473	0.0438	0.0579	40
17	f5-aa	relations	95.6181	41.3161	215.2174	40
17	f5-aa	verify	0.0019	0.0016	0.0026	40
17	f5-tail	factor_base	0.1371	0.1286	0.1576	39
17	f5-tail	linear_algebra	0.0091	0.0024	0.0172	39
17	f5-tail	oracle_setup	0.0492	0.0437	0.0581	39
17	f5-tail	relations	55.7400	23.7127	125.0042	39
17	f5-tail	verify	0.0020	0.0016	0.0027	39
17	f6-native	factor_base	0.1349	0.1291	0.1594	40
17	f6-native	linear_algebra	0.0100	0.0031	0.0198	40
17	f6-native	oracle_setup	0.0472	0.0434	0.0647	40
17	f6-native	relations	112.7684	45.7423	241.6172	40
17	f6-native	verify	0.0020	0.0016	0.0026	40
17	generalized-typed	factor_base	0.1356	0.1286	0.1576	40
17	generalized-typed	linear_algebra	0.0094	0.0026	0.0154	40
17	generalized-typed	oracle_setup	0.0467	0.0433	0.0565	40
17	generalized-typed	relations	91.7102	37.1233	197.4759	40
17	generalized-typed	verify	0.0020	0.0016	0.0027	40
17	semaev-batched	factor_base	0.1437	0.1393	0.1686	33
17	semaev-batched	linear_algebra	0.0071	0.0035	0.0104	33
17	semaev-batched	oracle_setup	0.0183	0.0156	0.0257	33
17	semaev-batched	relations	0.6191	0.2713	1.2083	33
17	semaev-batched	verify	0.0019	0.0016	0.0026	33
17	semaev-full-table	factor_base	0.1361	0.1309	0.1666	34
17	semaev-full-table	linear_algebra	0.0071	0.0026	0.0091	34
17	semaev-full-table	oracle_setup	0.0569	0.0552	0.0744	34
17	semaev-full-table	relations	0.5934	0.2899	1.2396	34
17	semaev-full-table	verify	0.0020	0.0016	0.0023	34
23	f4-baseline	factor_base	0.1538	0.0582	0.1826	40
23	f4-baseline	linear_algebra	0.0261	0.0048	0.0438	40
23	f4-baseline	oracle_setup	0.0767	0.0729	0.0878	40
23	f4-baseline	relations	2592.4235	246.9147	4210.3284	40
23	f4-baseline	verify	0.0036	0.0032	0.0050	40
23	f4-aa	factor_base	0.1557	0.0607	0.1963	40
23	f4-aa	linear_algebra	0.0276	0.0045	0.0415	40
23	f4-aa	oracle_setup	0.0761	0.0728	0.0996	40
23	f4-aa	relations	2593.6937	247.0205	4206.4476	40
23	f4-aa	verify	0.0037	0.0032	0.0049	40
23	f4-tail	factor_base	0.1561	0.0620	0.2481	40
23	f4-tail	linear_algebra	0.0232	0.0053	0.0417	40
23	f4-tail	oracle_setup	0.0771	0.0738	0.1517	40
23	f4-tail	relations	1636.0355	154.8664	2650.6260	40
23	f4-tail	verify	0.0037	0.0033	0.0049	40
23	f5-baseline	factor_base	0.1564	0.0594	0.1927	40

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
23	f5-baseline	linear_algebra	0.0255	0.0033	0.0438	40
23	f5-baseline	oracle_setup	0.0766	0.0731	0.0980	40
23	f5-baseline	relations	2763.9939	263.3492	4492.0319	40
23	f5-baseline	verify	0.0036	0.0032	0.0046	40
23	f5-aa	factor_base	0.1536	0.0593	0.1904	40
23	f5-aa	linear_algebra	0.0267	0.0056	0.0505	40
23	f5-aa	oracle_setup	0.0775	0.0726	0.0996	40
23	f5-aa	relations	2766.2142	263.4290	4493.2206	40
23	f5-aa	verify	0.0036	0.0032	0.0052	40
23	f5-tail	factor_base	0.1545	0.0618	0.1813	40
23	f5-tail	linear_algebra	0.0228	0.0028	0.0429	40
23	f5-tail	oracle_setup	0.0775	0.0722	0.0991	40
23	f5-tail	relations	1806.3969	170.4640	2928.3321	40
23	f5-tail	verify	0.0037	0.0032	0.0045	40
23	f6-native	factor_base	0.1563	0.0602	0.2034	40
23	f6-native	linear_algebra	0.0308	0.0041	0.0526	40
23	f6-native	oracle_setup	0.0757	0.0734	0.0953	40
23	f6-native	relations	3951.8285	374.9495	6444.7310	40
23	f6-native	verify	0.0037	0.0032	0.0048	40
23	generalized-typed	factor_base	0.1500	0.0567	0.1898	40
23	generalized-typed	linear_algebra	0.0264	0.0038	0.0478	40
23	generalized-typed	oracle_setup	0.0742	0.0706	0.0944	40
23	generalized-typed	relations	5945.8935	560.9768	9731.2949	40
23	generalized-typed	verify	0.0037	0.0033	0.0043	40
23	semaev-batched	factor_base	0.1623	0.1538	0.2572	39
23	semaev-batched	linear_algebra	0.0143	0.0023	0.0239	39
23	semaev-batched	oracle_setup	0.0315	0.0288	0.0379	39
23	semaev-batched	relations	20.3236	1.9198	33.8123	39
23	semaev-batched	verify	0.0037	0.0032	0.0044	39
23	semaev-full-table	factor_base	0.1497	0.0568	0.1804	39
23	semaev-full-table	linear_algebra	0.0150	0.0025	0.0228	39
23	semaev-full-table	oracle_setup	0.1197	0.1162	0.1407	39
23	semaev-full-table	relations	19.6966	1.9538	33.6351	39
23	semaev-full-table	verify	0.0036	0.0032	0.0047	39